

A reach map that moves: showing an operator the tilt-bounded workspace of a free-floating arm

[authors, order to agree]

Abstract

An arm on a free-floating spacecraft turns the spacecraft when it moves. If the base must stay within a tilt budget, here 2 degrees, only part of the arm’s reach can be used, and an operator needs to know which part. The obvious aid is a map of that part, drawn once. We show in simulation that such a map is wrong: the edge of the tilt-bounded reach depends on where the hand starts. Over 40 starts and 20 directions the edge stays within 5% of the map in 80% of cases and falls below 80% of it in 7%, down to 35% at worst, and we found no cheap rule that says which. This leaves three honest things to show an operator: a small map that held from every start we tested, a map recomputed from the current state that is right but 12 to 13 s late, or no map and a rehearsal that refuses a bad command. We describe the three, a tool that runs them, and a within-subject study protocol that compares them against the fixed map. [study results: no participant has been run]

1 Introduction

A manipulator on a free-floating base has no fixed workspace in the usual sense. Momentum is conserved, so every joint motion moves the base, and where the hand ends up depends on the path the joints took [1, 2]. Reaction null-space control [6, 7] removes the base motion when the arm has joints to spare, but only for part of the reach, and the rest costs base attitude.

In practice the attitude has a budget: an antenna has to stay pointed, a camera has to keep its target. So the question an operator actually has is not “can the arm reach that” but “can the arm reach that with the base still inside its budget”. The answer is a region, and the natural aid is to draw it.

This paper is about what happens when that region is not fixed. Our contributions:

1. A measurement of how the edge of the tilt-bounded reach moves with the start state, for a 7 joint arm on a free-floating bus, and two explanations that we tested and that failed.
2. An inner region that held from every start we tested, with its cost in reach, and a recomputed map with its cost in time.
3. A teleoperation tool and a pre-registered study design that compare what to show the operator. [and the result]

2 Related work

Workspaces of free-floating arms. Umetani and Yoshida [1] give the generalized Jacobian and already separate a workspace reachable along a straight path from a given posture from one that is guaranteed. Papadopoulos and Dubowsky [3] show that singularities depend on the path taken and

define path-dependent and path-independent workspaces. Vafa and Dubowsky [4] give the virtual manipulator. In all of these the workspace is bounded by singularity or by link lengths. None is bounded by a budget on base attitude.

Keeping the base still. Torres and Dubowsky [5] map which joint motions disturb the base least. Reaction null-space control [6] and its flight test on ETS-VII [7] remove the disturbance where redundancy allows. These are controllers. They do not tell an operator where the controller will stop working. [Sato 2002: slows the arm to keep attitude in limits, no user in the loop. Full reference to check against the paper itself]

Showing limits to operators. Virtual fixtures [8, 9] put limits in front of users, and have user studies, but the limits are fixed in space. [DLR null-space wall, arXiv 2001.10779: a haptic wall that moves with the state, simulation only, no users. Check the full reference] Predictive displays show where a delayed robot will be, not where it may go.

The gap. We found no work where the limit shown to the operator is a base attitude budget, where that limit is shown to move with the state, and where people are tested against it. This is what five searches turned up, not a proof that nobody did it. [one more systematic search before submission]

3 System

A free-floating bus carries a 7 joint arm in MuJoCo [10]. Nothing holds the bus: no thrusters, no reaction wheels. The hand is driven by resolved-rate control on the generalized Jacobian, and the spare joint is used to cancel base reaction where it can. Base tilt is counted from the attitude at the rest pose, and the budget is 2 degrees. [masses, link lengths, gains: table from the CA3 paper]

The reach is probed along 98 directions from the rest position of the hand. Along each one the hand is driven out until the tilt reaches the budget, a joint limit stops it, or the arm would hit the bus. The 98 end points form the map the operator sees, which we call the rest map. Distance along a direction is given as a fraction of the rest map, so 1 is its edge.

A command is checked by rehearsal: the move is simulated from the state the arm is really in, and refused if it ends outside the budget. Rehearsal is the only test in this paper that is always right, and it takes about 4 s for a far goal.

4 The edge moves

4.1 Measurement

We moved the hand to a random start one move away from rest, then found by bisection how far it could go along each of 20 random directions before the budget stopped it, and compared with the same search from rest. 40 starts, 800 start and direction pairs.

In 80% of pairs the edge is within 0.05 of the rest map. In 7% it is below 0.8, and the worst is 0.35 (Fig. 1, left). The bad pairs are not a corner of the space: 17 of the 40 starts and 13 of the 20 directions have at least one. They are more common on moves across the start direction (8 to 10% against 3 to 4% along it) and when the start left more tilt behind (15% above 0.8 degrees against 5% below 0.4), but neither tells you which pair will fail.

So a map drawn from rest is wrong for about one command in fourteen near its edge, and the operator cannot tell which one.

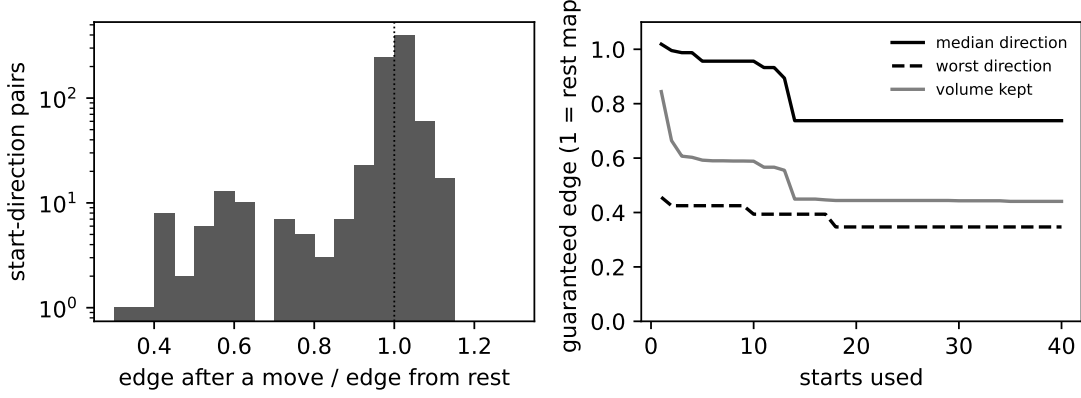


Figure 1: Left: edge after one move, as a fraction of the edge from rest, 800 start and direction pairs. Most are at 1, a tail collapses. Right: the edge that holds from every start used so far, against the number of starts. Scripted runs.

4.2 Two explanations that failed

Leftover tilt. The tilt is counted from rest, so a move that leaves tilt behind has less budget for the next one. If that were the cause, the right map would be the rest map with the budget corrected. We repeated the search with the leftover tilt forgiven. The edge did not go back: the median distance to the rest edge was 0.05 with tilt forgiven against 0.03 with it kept. Tilt is a vector and the budget is a ball around the rest attitude, so leftover tilt moves the edge outward in some directions and inward in others.

Enclosed area. For a free-floating system the attitude change over a closed path grows with the area the path encloses. Large shifts do only happen at large area (worst -0.29 m above 0.09 m², -0.05 m below 0.03 m²), but a linear fit on signed area explains 7% of the shift.

We do not have a formula for the edge. What we have is the simulation.

5 Three things to show instead

5.1 A map that held from every start

Take, for each direction, the worst edge over all starts. After 40 starts the median direction keeps 0.74 of the rest edge and the worst keeps 0.35, which is 44% of the rest reach by volume. The figure was the same at 20 starts as at 40 (Fig. 1, right), so for starts one move from rest it has probably stopped shrinking. It is not a guarantee, and a second move shows it. From 40 starts two moves away from rest, one had already spent the budget on the way. From 24 of the other 39 the rehearsal refused at least one tip of this map, 4.3% of all pairs, on 70 of the 98 directions. Pulling every direction in to 0.8 of its length leaves 11 starts and 0.9% of pairs. A fixed map cannot be made safe by shrinking it against sampled starts. Measured again on the 98 directions of the map itself, same 40 starts, the map that held keeps 0.67 of the rest edge on the median direction, 0.24 on the worst, and 40% of the volume. 28 of the 98 directions keep less than half and 5 keep everything.

5.2 A map from where the arm is

Run the 98 probes again from the current state. On one thread this takes 85 s. Shared over 10 processes it takes 12.6 s from rest and 7.4 s after a 0.3 m move, and 12 processes are no faster. So

this map is right, for a hand that has stopped, and 12 to 13 s old. The tool shows a yellow marker on the hand while the map is old and a green one when it is current.

5.3 No map

Draw nothing. Rehearse every command and refuse the ones that would break the budget. This is never wrong and gives the operator nothing to plan with: they find the edge by being refused.

6 Study

6.1 Design

Within subject, four conditions: the rest map (the control), the map that held, the map from the current state, and no map. 20 participants, balanced Latin square over four orders, five per order. Goals come in pairs. The first is easy and moves the arm off rest, and is not scored. The second is the test and starts from where the first left the arm. Whether a goal can be reached is rehearsed from the arm's real state when it appears. A participant either puts the hand on the goal or gives it up as out of reach, at a cost of 30 s.

Measures, on the second goal of each pair: seconds per goal with the penalty and the rehearsal time in it (primary), commands the fence had to bring back, reachable goals given up, unreachable goals given up, worst tilt, and after each block raw NASA-TLX [11] and one trust rating from 1 to 7.

Hypotheses, written before data: the rest map has the most commands brought back (H1); the map that held has none and the most reachable goals given up (H2); the current-state map is the fastest per goal (H3); no map has no commands brought back but is slower and rated as more work (H4). Friedman test over conditions on the primary measure, Wilcoxon pairs with Holm correction.

6.2 Results

[No participant has been run. Ethics approval and consent to settle first. A scripted operator will be run through all blocks before people are, to see whether seven scored goals per block can separate the conditions at all; that run will be reported as scripted, not as data.]

7 Limits

Simulation only. One arm, one bus mass, one budget. The map that held is tested on one-move starts. The delay of the current-state map is the delay on one laptop. The reach is probed on 98 lines from one point, so the map is a set of lines and not a volume with a proof behind it. No contact, no flexible parts, no sensor noise.

References

- [1] Y. Umetani and K. Yoshida, "Resolved motion rate control of space manipulators with generalized Jacobian matrix," *IEEE Trans. Robot. Autom.*, vol. 5, no. 3, 1989.
- [2] S. Dubowsky and E. Papadopoulos, "The kinematics, dynamics, and control of free-flying and free-floating space robotic systems," *IEEE Trans. Robot. Autom.*, vol. 9, no. 5, 1993.

- [3] E. Papadopoulos and S. Dubowsky, “Dynamic singularities in free-floating space manipulators,” *ASME J. Dyn. Syst. Meas. Control*, vol. 115, 1993.
- [4] Z. Vafa and S. Dubowsky, “The kinematics and dynamics of space manipulators: the virtual manipulator approach,” *Int. J. Robot. Res.*, vol. 9, no. 4, 1990.
- [5] M. A. Torres and S. Dubowsky, “Minimizing spacecraft attitude disturbances in space manipulator systems,” *J. Guid. Control Dyn.*, vol. 15, no. 4, 1992.
- [6] D. N. Nenchev, K. Yoshida, P. Vichitkulsawat, and M. Uchiyama, “Reaction null-space control of flexible structure mounted manipulator systems,” *IEEE Trans. Robot. Autom.*, vol. 15, no. 6, 1999.
- [7] K. Yoshida, K. Hashizume, and S. Abiko, “Zero reaction maneuver: flight validation with ETS-VII space robot and extension to kinematically redundant arm,” in *Proc. IEEE ICRA*, 2001.
- [8] L. B. Rosenberg, “Virtual fixtures: perceptual tools for telerobotic manipulation,” in *Proc. IEEE Virtual Reality Annu. Int. Symp.*, 1993.
- [9] J. J. Abbott, P. Marayong, and A. M. Okamura, “Haptic virtual fixtures for robot-assisted manipulation,” in *Robotics Research*, Springer, 2007.
- [10] E. Todorov, T. Erez, and Y. Tassa, “MuJoCo: a physics engine for model-based control,” in *Proc. IEEE/RSJ IROS*, 2012.
- [11] S. G. Hart and L. E. Staveland, “Development of NASA-TLX: results of empirical and theoretical research,” in *Human Mental Workload*, 1988.

[check every reference against the paper itself before submission: volume, issue, pages]